

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2025
 Programme: BE Full Marks : 100
 Course: Object Oriented Design and Modeling through UML (New) Pass Marks : 45
 Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is the role of notations in a modeling tool used for software design? Describe any five commonly used UML notations, explaining each with a brief note. 8
- b) Draw a Use Case Diagram for a Library Management System. Ensure to include relevant actors and use cases. Write the fully dressed use case format for the "Borrow Book" functionality in the Library Management System. 7
2. a) Differentiate between Association, Aggregation, and Composition in Object-Oriented Design. Illustrate these relationships using a Design Class Diagram (DCD) with an example such as a University Course Registration System. 7
- b) Define GRASP Patterns. Explain the Controller and Information Expert patterns with suitable examples. 8

OR

Discuss the role of Design Patterns in Object-Oriented Design. Explain the High Cohesion principle with an example scenario.

3. a) What is an operation contract in OOAD? List its essential parts and write a sample contract for a "processOrder" system operation in an e-commerce system. 7

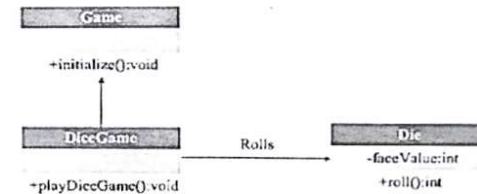
OR

Differentiate between Object-Oriented Analysis and Design (OOAD) and other software development paradigms such as Structured Analysis. What are the key advantages of using the Object-Oriented approach in system design? Support your answer with suitable examples.

- b) A company plans to develop an Online Event Ticket Booking System. Customers can create accounts, browse events by category, select seats, make payments, and receive digital tickets via email. They can cancel tickets before the event date. Admin users can

manage events, venues, pricing, and generate booking reports. The system must verify payments and update seat availability automatically.

- i. Draw a Use Case Diagram showing all actors and interactions.
 - ii. Draw a Sequence Diagram illustrating the process of a customer booking an event ticket and completing payment.
4. a) Define associations and attributes in a domain model. Explain how to identify conceptual classes or domain objects, and mention common mistakes in the identification process. 7
 - b) Design a Package Diagram for the Smart University Management System by organizing the system into key packages such as User Authentication, Course Management, Attendance Management, Resource Access, Notification Services, and System Logging. Show the dependencies between these packages and briefly explain the responsibility of each package in the system. 8
 5. a) Draw a System Sequence Diagram (SSD) for the use case 'Student Accesses Digital Resource' in the Smart University Management System for above scenario. 7
 - b) Convert the given Design Class Diagram into an equivalent code. (you can choose any language like java, c++, etc). 8



6. a) Explain the process of transforming Design Models into Implementation Code. Discuss how exception handling can be represented in UML diagrams. 7
- b) Compare and contrast Sequence Diagrams and Collaboration (Communication) Diagrams. Explain each with a suitable example, and discuss their respective strengths and weaknesses. 8
7. Write short notes on: (Any two) 2×5
 - a) Functional vs non- Functional Requirements
 - b) Rational Unified Process
 - c) Cohesion vs Coupling

POKHARA UNIVERSITY

Level: Bachelor

Semester – Spring

Year: 2020

Program: BE

Full Marks: 70

Course: Object Oriented Design and Modeling through UML

Pass Marks: 31.5

Time: 2 hrs.

Candidates are required to answer in their own words as far as practicable.

Answers with a valid set of UML diagrams indicate full marks.

Attempt all the questions.

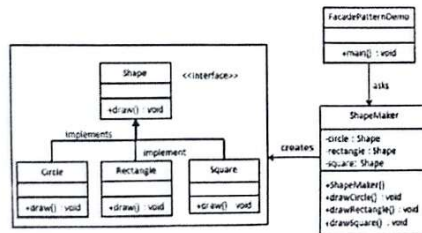
Group - A: (5×10=50)

- Q. N. 1 Discuss the emergence of OOA and OOD phases in addition to the existing OOP. Support your answer with key activities in each phase chronologically. 10
- Q. N. 2 Differentiate the structural and behavioral analysis process. Elaborate your understanding by using appropriate artifacts for both. 10
- Q. N. 3 Consider a business of your choice and develop a domain model. Follow the standard practices to find conceptual classes, attributes and associations to develop the domain model. 10
- Q. N. 4 Differentiate sequence diagram with collaboration diagram. Why Contracts are used and what are its format? 10

OR

What is visibility in OOAD? Describe its role in obtaining high abstraction. What are the various types of them? Elaborate using diagram and code snippets.

- Q. N. 5 Convert the diagram into code using any of the object oriented languages (C++, Java etc.) 10



Group - B: (1×20=20)

Q. N. 6

Case Study

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Due to a recent outbreak many sectors have been seriously affected. Face masks are mandatory in public places because it is a respiratory sensitive disease. A reputed hospital in Kathmandu wants to hire a software development team to develop a smart face mask detection system.

The system needs to be connected with several cameras for detecting human faces to make sure whether there is anyone without a face mask. The admin can add different wards with details including a supervisor and rooms with cameras in each ward. The supervisor can monitor own ward with a dashboard showing the total number of faces with mask and without masks. Additionally, there is a smart entrance gate that will block the people without a face mask.

Consider yourself as a team lead of the software development team. Illustrate your understanding about the context by developing and finalizing the following artifacts.

Use Case Diagram, Sequence Diagram, Class Diagram, Communication Diagram.

POKHARA UNIVERSITY

Level: Bachelor
 Programme: BE
 Course: Object Oriented Design and Modeling through UML

Semester: Spring
 Year : 2021
 Full Marks: 100
 Pass Marks: 45
 Time : 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
 The figures in the margin indicate full marks.
 Attempt all the questions.*

1. a) How does OOA differ from OOD? Elaborate your answer with key activities in each phase. 7
 b) What are use cases and how they can be used for identifying and recording system requirements? How would you differentiate between include and extend relationship between use cases? 8
2. a) What are different strategies for identifying conceptual classes? Explain with examples. 8
 b) What is an operation contract? Is it essential to develop an operation contract for each and every system operations? Explain. 7
3. a) What is visibility? Describe the use of attribute and parameter visibility among classes while designing class diagrams. 8
 b) Discuss the necessity of interaction diagram during software design. How is conditional messages and mutually exclusive conditional paths illustrated in interaction diagrams? Explain. 7
4. a) Explain different types of relationship used in class diagram. 8
 b) "UML is a language for specializing, visualizing, constructing and documenting the artifacts of the system." Justify this statement explaining the significance of UML in OOA/D. 7
5. a) What is the significance of studying exception and error handling in objects oriented analysis and design? Explain briefly about exception handling technique. 7
 b) Explain the process of mapping design to code (both class definition and method definition) with example. Support your answer by using a design class diagram and collaboration diagram. 8

6. Online Learning platforms help students to search for courses as per their criteria. If the system shows a suitable course then the students registers for the course and paying for it. Interaction with teacher can be done by registered students. Students need to attend the class and submit their assignments regularly. Students who attended 80% of the classes and submitted all their assignment are awarded with a course completion certificate at the end of the course. Also, the system facilitates the students to cancel their registration and refund the payment if it is done within a week. Now draw a use case model (no description required), Domain model, interaction diagrams and design class diagram from these information.

7. Write short notes on: (Any two)

2×5

- a) Functional Vs. Non-Functional Requirements
- b) Coupling and cohesion
- c) Reverse Engineering

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2022
Programme: B.E. Full Marks: 100
Course: Object Oriented Design and Modeling Pass Marks: 45
through UML Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Mention the significance of analysis and design in software development lifecycle from an object oriented perspective. 7
b) Define unified process with its advantages and disadvantages? 8
2. a) Case Study 8
A college canteen system has been planned to be implemented at XYZ College to improve the efficiency and convenience of purchasing food and drinks. The aim of the project was to streamline the process of ordering and payment while providing a better overall experience for the students. Students can order food items using a mobile app and get notified when the item is prepared. Chef uses a kitchen terminal to get orders, prepare the ordered food item and mark it as prepared after preparation.
Draw use case diagram above system and activity diagram for food ordering activity.
b) What do you understand by the term domain model? Explain components of a domain model. 7
3. a) "A conceptual model is the source of a design class diagram", elaborate this statement with an appropriate example. Also mention the steps involved. 8
b) How does a conceptual class differ from a software artifact? Discuss about the different methods for identifying conceptual classes. 7
4. a) What is a collaboration diagram? How do we represent mutually exclusive conditions in a collaboration diagram? Justify with an example. 8
b) Why design patterns are considered as important artifacts in object oriented design? Explain the use of Creator and Information Expert patterns. 7

5. a) What is visibility? Describe the use of attribute and parameter visibility among classes while designing class diagrams. 8
b) Draw system sequence diagram for enterItem in POS. Explain system events and system operation. 7
6. a) What is the significance of studying exception and error handling in objects oriented analysis and design? Explain briefly about exception handling technique. 7
b) Explain the process of mapping design to code (both class definition and method definition) with example. Support your answer by using a design class diagram and collaboration diagram. 8
7. Write short notes on: (Any two) 2×5
a) Functional Vs. Non-Functional Requirements
b) Operation contract
c) Representation of system behavior

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2023
Programme: BE Full Marks: 100
Course: Object Oriented Design and Modelling through Pass Marks: 45
UML Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) Explain the difference between Object-Oriented Analysis and Object-Oriented Design with an example. 7
b) What is Use case diagram? Explain include and extends relationship with example. Draw Use Case Diagram for Online Hotel Reservation System. 8
2. a) What is a contract? Explain different parts of contract with suitable example. 8
b) What is the purpose of developing a conceptual model? How is a domain model represented? Discuss its components. 7
3. a) What is the purpose of using system sequence diagram? Explain it with suitable example. 8
b) What is class diagram and why they are important in designing the software system? How are classes linked to each other? Show with different relationship used in class diagram with examples. 7
4. a) What is GRASP and list out all the patterns also provide the reason why design patterns are used in object oriented design? Explain the use of low coupling and high cohesion. 8
b) Discuss the scope and visibility of objects. Elaborate the use of parameter visibility and local visibility in details. 7
5. a) Explain the overall process of Object-Oriented Software Development (analysis, design and modelling, implementation). 8
b) Explain the process of mapping design to code (both class definition and method definition) with example. 7
6. a) Differentiate error and exceptions. How are exceptions handled in UML diagram? Explain. 7

- b) Why is it important to list the requirements before making the actual design? Explain with the help of Library Management System. 8
7. Write short notes on: (Any two) 2×5
a) Unified process
b) Sequence vs Collaboration diagram
c) Creating methods from collaboration diagram

POKHARA UNIVERSITY

Level: Bachelor Semester : Spring Year : 2023
Programme: BE Full Marks: 100
Course: Object Oriented Design and Modeling Pass Marks: 45
through UML Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Explain the difference between Object-Oriented Analysis and Object-Oriented Design with an example. 7
- b) Why is use case description important? Provide an example and discuss about the important section of a use case description. 8
- a) Discuss about the different phases of unified process and their role during software development. 8
- b) What is a contract? Explain different parts of contract with suitable example. 7
- a) What is the purpose of developing a conceptual model? How is a domain model represented? Discuss its components. 7
- b) "Designers make a bridge between customer requirements and the actual working code", Justify this statement using major sets of design artifacts. 8

Case Study

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Online Medical Store system allows us access their system with a valid mobile number. We can search for medicines in their systems by providing the doctor's prescription list. The prescription list is compared and a list of available medicines is displayed to us. Then we select the medicines and proceed to checkout and make payment. We also indicate the address for delivery. The system displays the expected delivery date. After that, a SMS with the delivery details is sent to our mobile number. Also, the system facilitates the customers to cancel their order 1 week before delivery with no cancellation charge. After 1 week no cancellation can be made.

Draw a use case model, Domain model, collaboration diagrams and design class diagram from these information.

- 5. a) Discuss the scope and visibility of objects. Elaborate the use of parameter visibility and local visibility in details 7
- b) How cohesion and coupling affect a good design? Explain with suitable points. 8
- 6. a) Explain about Fault, error and failure with relevant examples. Discuss the steps to handle exception in object oriented programming. 8
- b) Designs are for programmers not for end users. Show how different design diagrams help programmers for writing code. 7
- 7. Write short notes on: (Any two) 2×5
 - a) Conceptual class category list
 - b) Collection Classes
 - c) Information expert

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year : 2024

Programme: BE

Full Marks : 100

Course: Object Oriented Design and Modeling through UML

Pass Marks : 45

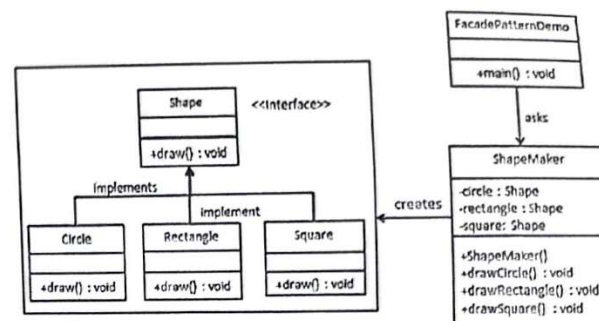
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) What is the unified process? Explain its phases and highlight how each contributes to software development. 8
- b) Explain the concept of a "contract" in object-oriented systems. Discuss its components with an appropriate example. 7
2. a) Design a system for an online movie ticket booking platform. The system must allow searching for movies, viewing showtimes, booking tickets, and making payments. The system should also allow managing user accounts, movie catalogs, and payment security. Based on these requirements, create the following: 8
 - Use Case Diagram
 - Sequence Diagram
- b) Define design patterns. Discuss how the design principle of encapsulation supports both low coupling and high cohesion in object-oriented systems. 7
3. a) Describe the concept of inheritance in object-oriented design. Illustrate with a UML class diagram how inheritance can be used to model a hierarchy of classes in a system. 8
- b) How is exception handling represented in UML? Provide an example using an activity diagram or sequence diagram to illustrate exception handling in a system. 7
4. a) Implement the following design into code using any OOP languages (C++, Java). 8



- b) How does the Object-Oriented Analysis and Design paradigm differ from other paradigms? Explain the distinction between Object-Oriented Analysis and Object-Oriented Design using an illustrative example. 7
5. a) What are associations and attributes in domain model. Explain how do you identify the domain objects or conceptual classes in Domain model including common mistakes that are made while identifying the conceptual classes. 7
- b) Define visibility. Describe the use of parameter and attribute visibility among classes while designing class diagrams. 8
6. a) Explain steps of mapping design to code. Discuss about exception handling in UML diagrams. 7
- b) **Cheque Payment:** 8

A customer presents a cheque to the bank employee. The employee checks if the cheque and the signature are valid and that the customer has enough amount of balance in the account. The maximum amount of withdrawal allowed is 100000 per cheque unless the customer is a Privileged Customer in which case the customer can make a withdrawal of 500000 per cheque. Upon successful verification, the system records the successful completion of the transaction and the cash is handed over to the customer.

 - i. Perform domain modeling of the above scenario.
 - ii. Draw the sequence diagram for the events that occur in the scenario.
7. Write short notes on: (Any two) 2×5
 - a) Functional Vs Non-Functional Requirement
 - b) Activity Diagrams
 - c) Rational unified process

POKHARA UNIVERSITY

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Semester: Spring
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 Full Marks: 100
 Pass Marks: 45
 Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) Explain the differences between Object-Oriented Analysis and Object-Oriented Design with an example. 7
 b) What is a contract? Explain different parts of a contract with a suitable example. 8
2. a) You are tasked with designing a system for an online bookstore. The primary features of the system include searching for books, viewing book details, adding books to a shopping cart, and proceeding to checkout. The system needs to handle user accounts, manage a catalog of books, and process payments securely. Based on the above requirements, design the following: 8
 i. Use Case Diagram
 ii. Activity Diagram
 b) Describe the concept of inheritance in object-oriented design. Illustrate with a UML class diagram how inheritance can be used to model a hierarchy of classes in a system. 7
3. a) Case Study: A startup aims to develop a platform where users can rent and share books within their city. Users can list their books, search for available books, and rent them. Administrators manage user accounts and oversee transactions. Draw a use case diagram for the system and an activity diagram for the book renting process. 8
 b) Draw system sequence diagram for purchase-ticket from Ticket Vending Machine. Explain system events and system operation. 7
4. a) Justify the statement: "Designers bridge the gap between customer requirements and the actual working code" by discussing the major sets of design artifacts. 8

- b) Explain the relationship between communication diagrams and sequence diagrams. Develop a communication diagram for a user login process and convert it to a sequence diagram. 7
5. a) What is the use of design patterns in object oriented design? Explain the use of low coupling and high cohesion. 8
 b) Discuss the scope and visibility of objects in UML. Elaborate on parameter visibility and local visibility with detailed examples. 7
6. a) "Designs are primarily for programmers, not end users." Discuss how different design diagrams assist programmers in writing code. 7
 b) Non-functional requirements in OOA can impose constraints in design and lead to exceptions in implementation. Explain methods for handling exceptions with relevant examples. 8
7. Write short notes on: (Any two) 2×5
 a) Functional Vs Non Functional Requirement
 b) Guard
 c) Swimlane

POKHARA UNIVERSITY

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 Time : 3 hrs.

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Attempt all the questions.

1. a) Justify the statement "OOA focuses on 'WHAT' and OOD focuses on 'HOW' with an example related to a mobile banking system" 7
- b) Design a system for a Food Delivery System with features like User Registration, Browsing Restaurants, Placing Orders and Tracking Delivery. Based on these requirements, create the following: 8
 - i. Use Case Diagram
 - ii. System Sequence Diagram
2. a) Define design patterns. Discuss how the design principle of encapsulation supports both low coupling and high cohesion in object-oriented systems. 7
- b) What do you understand by the Rational Unified Process? Describe phases and disciplines of Rational Unified Process. 8
3. a) What is visibility of object? Explain different types of visibility between objects? Illustrate with example. 8
- b) Define a Use Case and explain how use cases help in identifying and recording system requirements in OOAD. How do "include" and "extend" relationships differ in a use case diagram? Illustrate with examples. 7
4. a) How the following are represented in communication diagram. 8
 - i. Message to self or this
 - ii. Mutually exclusive conditional path
 - iii. Iteration over the collection
 - iv. Conditional message
 - v. Polymorphic message
- b) What are associations and attributes in domain model. Explain how do you identify the domain Objects or conceptual class in Domain

model including Common mistakes that are made while identifying the conceptual classes.

5. a) What is the significance of studying exception and error handling in object oriented analysis and design? Explain briefly about exception handling technique in UML. 8
- b) What are system events and system operations? What is a contract? Explain the format of writing a contract and provide a contract for the system operation to add a product to the cart in an E-commerce System. 7
6. a) What is the difference between a domain model and a class diagram? A company is planning to develop an Online Shopping System to handle product listings, customer orders, payments, and deliveries. The system should allow customers to browse products, add them to their cart, place orders, and make payments. Vendors should be able to manage product listings, view customer orders, and update order statuses. Delivery personnel should be able to view and update delivery statuses. Draw a class diagram for the Online Shopping System. Identify the main entities, their attributes, and the relationships between them. The diagram should clearly represent the key concepts and their associations within the system. 8
- b) What are the necessary steps to convert design to code? Explain how class definition and method definitions are implemented in code. 7
7. Write short notes on: (Any two) 2×5
 - a) Combined Fragments
 - b) Design Class Diagram
 - c) Creating methods from interaction diagram

POKHARA UNIVERSITY

Level: Bachelor
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Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Mention the significance of analysis and design in software development lifecycle from an object oriented perspective. 7
- b) "UML is a language for specializing, visualizing, constructing and documenting the artifacts of the system." Justify this statement explaining the significance of UML in OO/AD. 8
2. Case Study: Bank Transaction System:

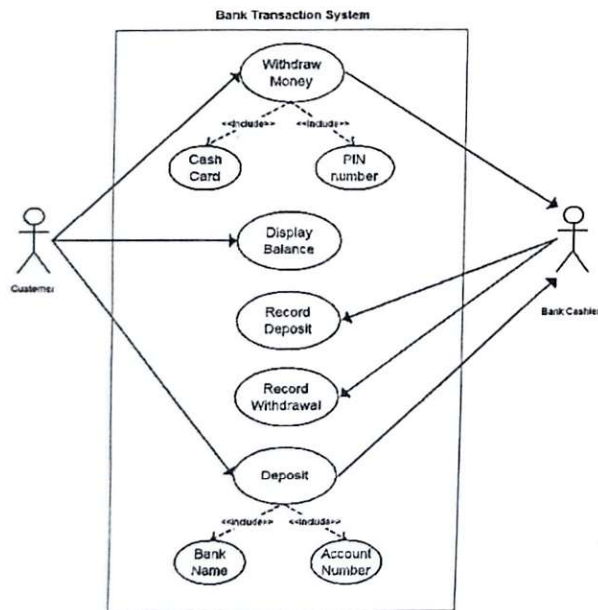


Fig: use case of Bank Transaction System

On the basis of above Use case:

- a) Develop the system sequence diagram for Withdraw Money and Deposit functions. 7
- b) Write the contract for any one of the system operations you identify. 8
3. a) What is the purpose of developing a conceptual model? How is a domain model represented? Discuss its components. 7
- b) Describe the concept of inheritance in object-oriented design. Illustrate with a UML class diagram how inheritance can be used to model a hierarchy of classes in a system. 8
4. a) What is visibility? Describe the use of attribute and parameter visibility among classes while designing class diagrams. 7
- b) What are the roles of objects and patterns in OODM? Describe the different patterns used. 8
5. a) How cohesion and coupling affect a good design? Explain with suitable points. 7
- b) Justify the statement: "Designers bridge the gap between customer requirements and the actual working code" by discussing the major sets of design artifacts. 8
6. a) Explain about Fault, error and failure with relevant examples. Discuss the steps to handle exception in object oriented programming. 7
- b) What are descriptor classes in GRASP? How does it achieve singleton functionality? 8
7. Write short notes on: (Any two) 2×5
 - a) Unified process model
 - b) MVC Pattern
 - c) Mapping design to code